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COURSE: BACHELOR OF PROCUREMENT AND LOGISTIC MANAGEMENT

COURSE UNIT: ELEMENTS OF MATHS

QN 1

Solution

Let N, M, and B be the sets of people who read new vision, Monitor and bukedde respectively.

Let n(N), n(M) and n(B) be the number of people in each set.

n(N)=40

n(M)=35

n(B)=60

n(NnM)=7

n(MnB)=10

n(NnB)=4

n(NuMuB)'=34

Total population, n(u)=150

The total number of people is the sum of those who read at least one newspaper on those who read None.

n(NuMuB)

Let X be the number of people who read all three newspapers, i.e.

X = n(NnMnB)

n(NuMuB) = n(u)-n(NuMuB)'

n(NuMuB) = 150 – 34

=116

Formula for the union of three sets.

n(NuMuB) = n(N) + n(M) + n(B) – n(NnM) – n(MnB) -n(NnB) + n(NnMnB)

substitute the known values

116 = 40+35+60-7-10-4+X

116 = 114+X

X = 116-114

X = 2

a) The number of people who read all the three newspapers is 2

b) Number of people who read only one newspaper

Only N = 40 - (5+2+2) = 40 – 9 = 31

Only M = 35 – (5+8+2) = 35 – 15 = 20

Only B = 60 – (2+8+2) = 60 – 12 = 48

The number of people who read only one newspaper is the sum of these values.

Total only one newspaper = 31+20+48 = 99

The number of people who read only one newspaper is 99

c) Number of people who read N and M only

n(NnM) only = n(NnM) – n(NnMnB)

n(NnM) only = 7 – 2 = 5

The number of people who read N and M only is 5

d) Probability that a person chosen at random from this town reads two newspapers

Only N and M: 7 – 2 = 5

Only M and B: 10 – 2 = 8

Only N and B: 4 – 2 = 2

Total who read exactly two newspapers =

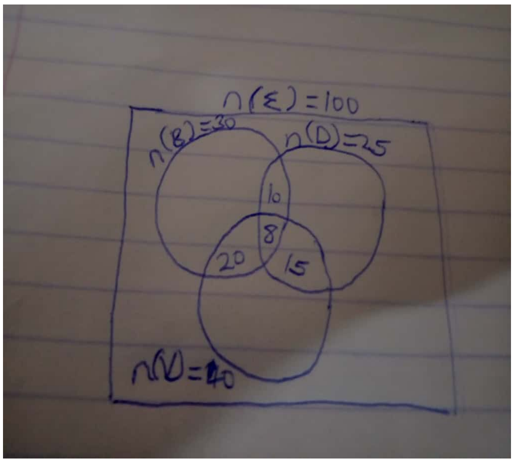
5+8+2 = 15

P (reads two newspapers only) = 15/150

= 1/10

QN2

VENN DIAGRAM



a) n(DnV) only = n(DnV) – n(BnDnV)

= 15 – 8 = 7

There are 7 students who played volley ball and badminton but not basket ball

b) n (B only) = 8

There are 8 students who played basketball only

c) n(BuDuV) = 8+8+13+2+7+12+8

= 58

Total students – students who played at least one sport

= 100 – 58

= 42

The number of students who did not play any of the sports is 42